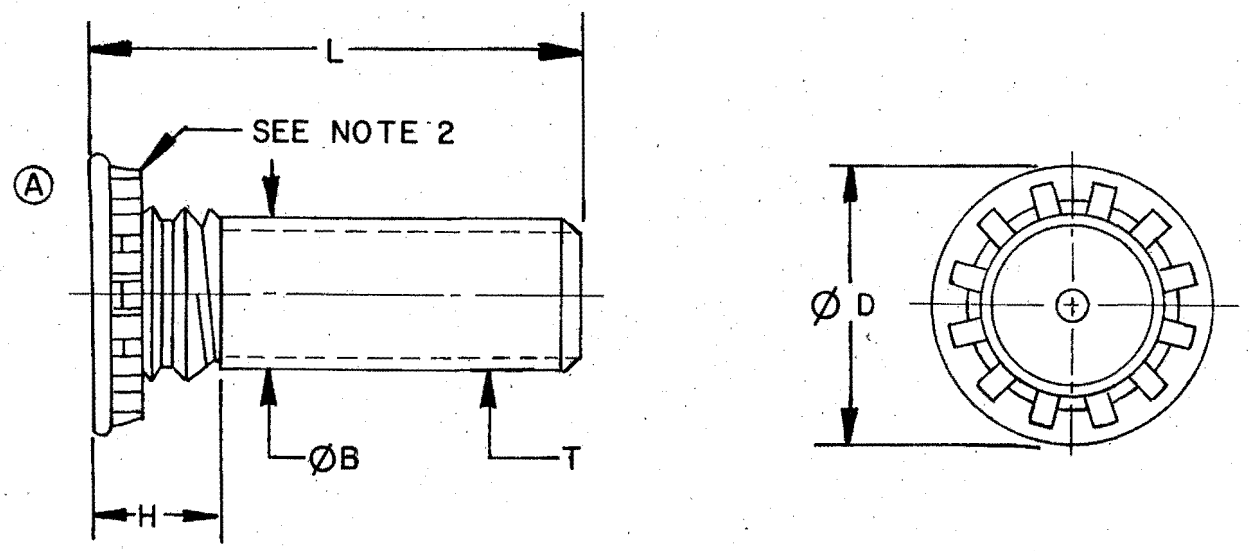
MILITARY SPECIFICATION SHEET

STUD, SELF-LOCKING, FLUSH HEAD, RIBBED CLINCH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the studs described herein shall consist of this specification and the latest issue of MIL-S-63540.



NOTES:

1. All dimensions are in inches.
Ⓐ 2. Clinching profile may vary.

FIGURE 1. STUD, SELF-LOCKING, FLUSH HEAD, RIBBED CLINCH TYPE.

Ⓐ Denotes changes

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TABLE I. PART DIMENSIONS

T THREAD SIZE CLASS 2A	Ø B STUD DIA	Ø D (A) ±.015	H (A) MAX
.086-56 UNC (2-56)	.086	.144	.075
.112-40 UNC (4-40)	.112	.176	.085
.138-32 UNC (6-32)	.138	.206	.090
.164-32 UNC (8-32)	.164	.237	.095
.190-24 UNC (10-24)	.190	.256	.105
.190-32 UNF (10-32)	.190	.256	.105
.250-20 UNC (1/4-20)	.250	.337	.125
.313-18 UNC (5/16-18)	.313	.376	.150

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TABLE II. THREAD SIZE AND LENGTH DASH NUMBERS

L ±.015	MATERIAL	T THREAD SIZE CLASS 2A							
		.086-56 UNC	.112-40 UNC	.138-32 UNC	.164-32 UNC	.190-24 UNC	.190-32 UNF	.250-20 UNC	.313-18 UNC
		THREAD SIZE AND LENGTH DASH NUMBER							
.250	CARBON STEEL CRES	1 1C	6 6C						
.313	CARBON STEEL CRES	2 2C	7 7C	12 12C					
.375	CARBON STEEL CRES	3 3C	8 8C	13 13C	20 20C	28 28C	36 36C		
.500	CARBON STEEL CRES	4 4C	9 9C	14 14C	21 21C	29 29C	37 37C	44 44C	
.625	CARBON STEEL CRES	5 5C	10 10C	15 15C	22 22C	30 30C	38 38C	45 45C	51 51C
.750	CARBON STEEL CRES		11 11C	16 16C	23 23C	31 31C	39 39C	46 46C	52 52C
.875	CARBON STEEL CRES			17 17C	24 24C	32 32C	40 40C	47 47C	53 53C
1.000	CARBON STEEL CRES			18 18C	25 25C	33 33C	41 41C	48 48C	54 54C
1.250	CARBON STEEL CRES			19 19C	26 26C	34 34C	42 42C	49 49C	55 55C
1.500	CARBON STEEL CRES				27 27C	35 35C	43 43C	50 50C	56 56C

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REQUIREMENTS:

Ⓐ 1. Material.

a. Carbon steel shall be Grade 1005 (UNS G10050) thru Grade 1018 (UNS G10180) in accordance with ASTM A 29 or ASTM A 510.

b. Corrosion resistant steel shall be Type XM-7 (UNS S30430) in accordance with ASTM A 493 or Types 303 (UNS S30300) and 303 Se (UNS S30323) in accordance with ASTM A 582.

2. Protective finish. Carbon steel studs shall be plated with electrodeposited cadmium in accordance with QQ-P-416, Type II, Class 3.

3. Surface treatment. Corrosion resistant steel studs shall be cleaned, descaled and passivated in accordance with ANSI/ASTM A 380.

Ⓐ 4. Hardness.

a. Carbon steel studs, after carburizing and tempering or carbonitriding, shall have a hardness of 75-87HR15N. The minimum effective depth of case shall be .010 inch.

b. Corrosion resistant steel studs shall have a hardness of 171HB minimum.

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5. Push-out and torque-out values. When the stud is properly installed (see installation procedures), push-out and torque-out values shall be as specified in Table III.

TABLE III. PUSH-OUT AND TORQUE-OUT VALUES.

T THREAD SIZE CLASS 2A	SHEET THICK-NESS (REF) 1/	PUSH-OUT LOAD (LBS) MIN 2/	TORQUE-OUT LOAD (IN LBS) MIN 2/	PUSH-OUT LOAD (LBS) MIN 2/	TORQUE-OUT LOAD (IN LBS) MIN 2/
		ALUMINUM ALLOY SHEET MATERIAL 3/		COLD ROLLED STEEL SHEET MATERIAL 3/	
.086-56 UNC	.040	70	4	120	4
	.060	115	4.5	180	4.5
.112-40 UNC	.040	85	9	160	9
	.060	160	9	220	9
.138-32 UNC	.040	100	16	165	20
	.060	175	16	280	24
.164-32 UNC	.040	120	30	175	36
	.060	230	32	320	36
.190-24 UNC	.040	150	36	200	48
	.060	280	40	400	48
.190-32 UNF	.040	150	36	200	48
	.060	280	40	400	48
.250-20 UNC	.062	240	80	320	100
.313-18 UNC	.093	400	140	650	160

1/ Test panel thickness (.040 or .060) is optional between thread sizes .086-56 UNC and .190-32 UNF, provided the corresponding load is maintained.
2/ The load values shown are the minimum non-destruct values that apply during testing.
3/ Sheet material hardness values shall not exceed 80HRB for use with carbon steel studs or exceed 70HRB for use with corrosion resistant steel studs.

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6. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table II.

Example: M63540/1-1C

- Dash number
- Specification sheet number

M63540/1-1C indicates - Stud, self-locking, flush head, ribbed clinch type; CRES; .086-56 UNC-2A thread; .250 length; .144 head diameter; .075 head height.

INSTALLATION PROCEDURES:

a. Any sheet thickness above the minimum indicated on Table IV for a specific thread size may be used.
b. Drill or punch hole in sheet material with diameter as shown in Table IV.
b. Locate shank of stud squarely in the hole in the mounting surface, with anvil below sheet (see Figure 2).
c. Apply force to embed the head of the stud flush with the sheet.

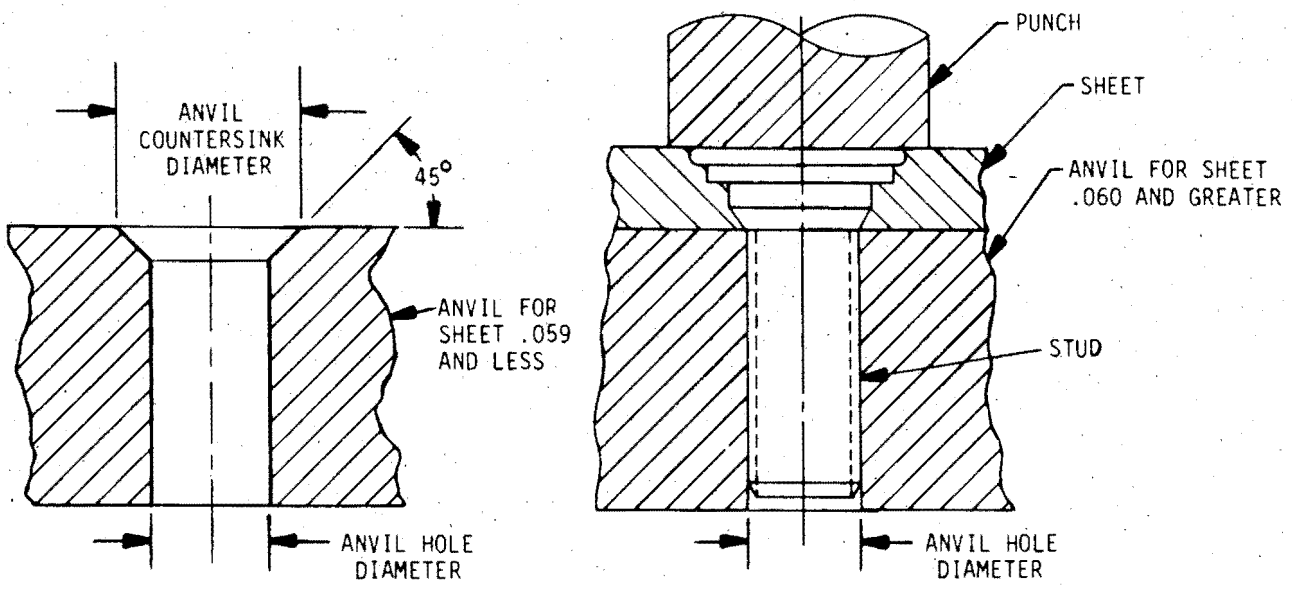


FIGURE 2. GENERAL TOOLING SETUP.

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TABLE IV. INSTALLATION DIMENSIONS

T THREAD SIZE CLASS 2A	HOLE SIZE IN SHEET +.003 -.000 (REF)	Ⓐ CENTER OF HOLE TO SHEET EDGE MIN (REF)	Ⓐ NUT TIGHTENING TORQUE (IN LBS) MAX (REF) 1/	ANVIL COUNTERSINK DIAMETER (REF)	ANVIL HOLE DIAMETER (REF)
.086-56 UNC	.085	.111	2.3	.110 -.114	.087 -.090
.112-40 UNC	.111	.219	5	.136 -.140	.113 -.116
.138-32 UNC	.137	.250	9	.162 -.166	.139 -.142
.164-32 UNC	.163	.281	17	.188 -.192	.165 -.168
.190-24 UNC	.189	.281	24	.210 -.220	.191 -.194
.190-32 UNF	.189	.281	27	.210 -.220	.250 -.253
.250-20 UNC	.249	.313	58		.250 -.253
.313-18 UNC	.311	.375	120		.313 -.316

1/ This load is the limiting load when the nut is fastened to the stud at installation.

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Custodians:
Army - AR
Navy - MC
Air Force - 99

Review activities:
Army - AT, ME
Air Force - 11, 82
DLA - IS

User activities:
Army - ER
Navy - OS, SH
NSA - NS

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5307-0332)

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